Final Scratch Work - Biostatistics

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Q5 Simple Linear Regression Question (A,B,C,D) 10 pts

a. **Fitted Equation**

The Fitted equation from this lienar model is:

$$\widehat{\text{CognitiveScoreChange}} = -5.160 - 0.148 \cdot \text{Age}$$

Where:

- **The Intercept ($\beta_0 = -5.160$)** is the predicted change in cognitive score when Age = 0. While this value is not meaningful in practice, it serves as our y-intercept in the linear model.

and:

- **The Slope ($\beta_1 = -0.148$)** indicates that each additional year of age is associated with a predicted decrease of 0.148 points in cognitive score.

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(broom)
```

```
Warning: package 'broom' was built under R version 4.3.3
```

```
library(gt)
```

```
Warning: package 'gt' was built under R version 4.3.3
```

Q2 Odds ratio and Chi-squared tests (A,B,C,D,E)

Q2(a): recreate table and compute OR & RR

```r
# Q2(a) manual calculations: OR, RR, 95% CIs

# Observed 2x2 (rows: drug1, drug2; cols: <7, >=7)
tab <- matrix(c(110, 46,
                74, 114),
              nrow = 2, byrow = TRUE,
              dimnames = list(Drug = c("drug1","drug2"),
                              A1c  = c("<7", ">=7")))
tab
```

```
       A1c
Drug     <7 >=7
  drug1 110  46
  drug2  74 114
```

```r
a <- tab[1,1]; b <- tab[1,2]
c <- tab[2,1]; d <- tab[2,2]

# Odds Ratio
OR <- (a*d)/(b*c)
logOR <- log(OR)
se_logOR <- sqrt(1/a + 1/b + 1/c + 1/d)
CI_OR <- exp(logOR + c(-1,1)*1.96*se_logOR)

# Risk Ratio
p1 <- a/(a+b)
p0 <- c/(c+d)
RR <- p1/p0
# Wald SE for log(RR)
se_logRR <- sqrt( (1/a) - (1/(a+b)) + (1/c) - (1/(c+d)) )
CI_RR <- exp(log(RR) + c(-1,1)*1.96*se_logRR)

# Pretty output
out <- data.frame(
  Measure = c("Odds Ratio (drug1/drug2)", "Risk Ratio (drug1/drug2)"),
  Estimate = c(OR, RR),
  Lower95CI = c(CI_OR[1], CI_RR[1]),
  Upper95CI = c(CI_OR[2], CI_RR[2])
)
out |> gt() |> tab_header(title = "Q2(a) Manual effect estimates with 95% CIs")
```

Q2(a) Manual effect estimates with 95% CIs			
Measure	Estimate	Lower95CI	Upper95CI
Odds Ratio (drug1/drug2)	3.683901	2.344929	5.787437
Risk Ratio (drug1/drug2)	1.791407	1.460247	2.197668

Q2 (d)

Under the null hypothesis of no association between drug regimen and achieving A1c < 7,
we assume the row and column variables are **independent**.
In that case, the probability of being in a given cell is the product of its row probability and column probability.
Multiplying these probabilities by the total sample size N gives the expected count formula:

$$E_{ij} = \frac{(\text{row total}_i) \times (\text{column total}_j)}{N}$$

For example:

-

$$E_{\text{drug1},<7} = \frac{156 \times 184}{344} = 83.48$$

-

$$E_{\text{drug1},\geq 7} = \frac{156 \times 160}{344} = 72.52$$

-

$$E_{\text{drug2},<7} = \frac{188 \times 184}{344} = 100.52$$

-

$$E_{\text{drug2},\geq 7} = \frac{188 \times 160}{344} = 87.48$$

```
# Q2(d): Expected counts under H0 (no association)

library(gt)

# Observed table (rows: drug1, drug2; cols: <7, >=7)
```

```r
O <- matrix(c(110, 46,
              74, 114),
            nrow = 2, byrow = TRUE,
            dimnames = list(Drug = c("drug1","drug2"),
                            A1c  = c("<7", ">=7")))

# Totals
row_tot <- rowSums(O)
col_tot <- colSums(O)
N <- sum(O)

# Expected counts formula: E_ij = (row_i * col_j) / N
E <- outer(row_tot, col_tot) / N

# Show the math in-line for clarity
explain <- sprintf(
  paste0("E[drug1,<7] = 156*184/344 = %.2f;  ",
         "E[drug1,>=7] = 156*160/344 = %.2f;  ",
         "E[drug2,<7] = 188*184/344 = %.2f;  ",
         "E[drug2,>=7] = 188*160/344 = %.2f."),
  E[1,1], E[1,2], E[2,1], E[2,2]
)

# Table of expected counts (rounded for display only)
make_table <- function(m, title){
  as.data.frame.matrix(round(m, 2)) |>
    tibble::rownames_to_column("Drug") |>
    gt() |>
    tab_header(title = title)
}


make_table(E, "Q2(d) Expected counts under H0")
```

Q2(d) Expected counts under H0		
Drug	<7	>=7
drug1	83.44	72.56
drug2	100.56	87.44

Q2(e): Chi-square statistic

We test independence using the Pearson chi-square:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \quad \text{df} = (r-1)(c-1) = 1.$$

```
# Recreate observed counts (rows: drug1, drug2; cols: <7, >=7)
O <- matrix(c(110, 46,
              74, 114),
            nrow = 2, byrow = TRUE,
            dimnames = list(Drug = c("drug1","drug2"),
                            A1c  = c("<7", ">=7")))

# Expected counts from part (d)
row_tot <- rowSums(O); col_tot <- colSums(O); N <- sum(O)
E <- outer(row_tot, col_tot) / N

# Chi-square components and statistic
contrib <- (O - E)^2 / E
chisq_stat <- sum(contrib)
df <- (nrow(O)-1)*(ncol(O)-1)
pval <- 1 - pchisq(chisq_stat, df = df)

# Nicely show contributions (optional)
as.data.frame.matrix(round(contrib, 3)) |>
  tibble::rownames_to_column("Drug") |>
  gt() |>
  tab_header(title = "Cell-wise chi-square contributions")
```

Cell-wise chi-square contributions

Drug	<7	>=7
drug1	8.453	9.721
drug2	7.014	8.066

```
# Final result
data.frame(Statistic = round(chisq_stat, 2), df = df, p_value = signif(pval, 4)) |>
  gt() |>
  tab_header(title = "Q2(e) Chi-square test result")
```

Q2(e) Chi-square test result

Statistic	df	p_value
33.25	1	8.085e-09

Q7 Lassa dataset

1) Pilot Data Summary

Outcome	n	mean_CT	sd_CT	median_CT	mean_age
Died	4	25.50	4.65	25.5	33.13
Survived	21	31.81	4.24	31.0	31.23

Interpretation: Survivors had higher CT values (lower viral load) than those who died. The group sizes are small and unequal, which guided the test selection in the next step.

2) Mean CT Difference (Unadjusted & Adjusted)

Comparison	Estimate	CI_Lower	CI_Upper	p_value
Unadjusted (Welch t-test)	-6.31	-13.26	0.64	0.065
Adjusted (Linear regression: +Age +Gender)	7.54	3.26	11.82	0.001

Methodology:
The Welch's t-test was used for the unadjusted comparison because it compares the means of two independent groups without assuming equal variances — a safer choice given our small and unequal group sizes. Linear regression was then used for the adjusted analysis to estimate the mean CT difference while controlling for age and sex, reducing confounding and isolating the association with survival.

Interpretation: The unadjusted result is borderline significant at α = 0.05. After adjustment, the mean CT difference is statistically significant, supporting the hypothesized relationship.

3) Required Sample Size for 90% Power

Formula:

$$n_{\text{per group}} = 2\left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{\Delta/s}\right)^2$$

- $z_{1-\alpha/2} = 1.96$

  for α = 0.05 (two-sided)

- $z_{1-\beta} = 1.2816$

for 90% power

- $s = 4.655$

  (larger pilot SD for conservatism)

- $\Delta = 4$

  (clinically meaningful mean difference)

Outcome	sd_CT
Died	4.655
Survived	4.238
SD used	4.655
Required n per group (Δ = 4)	29

Interpretation: Using the larger group SD makes the estimate conservative, ensuring adequate sample size even in the more variable group. Enrolling 29 per group will provide 90% power to detect a 4-point difference in CT at α = 0.05.

Conclusion

Pilot data suggest survivors present with higher CT values, indicating lower viral load at admission. Adjustment for age and sex strengthens the statistical evidence. The sample size estimate of 29 per group offers a practical target for designing the larger study, though results should be interpreted cautiously due to the small pilot sample.

Appendix: Full R Code

```
library(dplyr); library(broom)

dat <- read.csv("lassa.csv") |>
  mutate(
    Outcome = factor(ifelse(Died == 1, "Died", "Survived")),
    Gender  = factor(ifelse(Female == 1, "Female", "Male"))
  )

# 1) Summaries
summ_tab <- dat |>
  group_by(Outcome) |>
  summarise(
```

```
    n         = n(),
    mean_CT   = mean(CT, na.rm = TRUE),
    sd_CT     = sd(CT,   na.rm = TRUE),
    median_CT = median(CT, na.rm = TRUE),
    mean_age  = mean(Age, na.rm = TRUE),
    .groups = "drop"
  )

# 2) Unadjusted & adjusted
tt <- t.test(CT ~ Outcome, data = dat, var.equal = FALSE)
tt_row <- tidy(tt) |>
  transmute(Comparison = "Unadjusted (Welch t-test)",
            Estimate = estimate, CI_Lower = conf.low, CI_Upper = conf.high, p_value = p.v

fit <- lm(CT ~ Outcome + Age + Gender, data = dat)
adj_row <- tidy(fit, conf.int = TRUE) |>
  filter(term == "OutcomeSurvived") |>
  transmute(Comparison = "Adjusted (Linear regression: +Age +Gender)",
            Estimate = estimate, CI_Lower = conf.low, CI_Upper = conf.high, p_value = p.v

# 3) Sample size calc
sds <- dat |>
  group_by(Outcome) |>
  summarise(sd_CT = sd(CT, na.rm = TRUE), .groups = "drop")
s_used <- max(sds$sd_CT)
alpha <- 0.05; z_alpha <- qnorm(1 - alpha/2); z_beta <- qnorm(0.90); Delta <- 4
n_per_group <- ceiling(2 * ((z_alpha + z_beta)^2) * (s_used^2) / (Delta^2))
```